ASSD: Feature Aligned Single-Shot Detection for Multiscale Objects in Aerial Imagery

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***Abstract*— Object detection is a fundamental part of the interpretation of remote sensing imagery. The one-stage object detector has been adopted into this field because of its high computational efficiency. However, this detector suffers from the misalignment among predefined anchor, object, and feature extracted by standard convolution kernel both in spatial and scale. It limits the further improvement of performance, especially for the long-narrow and multiscale geospatial objects. In this article, the problem is defined as *the feature misalignment* problem. To deal with this issue, an efficient feature aligned single-shot detector (ASSD) is proposed, which consists of two modules: a novel pseudo anchor proposal module (PAPM) and a flexible context-based feature alignment module (CFAM). The PAPM replaces the regular anchor group with the proposed core anchor and refines it to get aligned locations. It can tackle the spatial misalignment between anchors and their corresponding objects and alleviate the negative/positive imbalance problem. Then, the CFAM adaptively adjusts the sampling points of the convolution kernel and collects the context information according to the aligned core anchor. This plug-and-play module can effectively rectify the misalignment between kernel and objects and extract aligned and robust features. A series of comprehensive experiments are conducted on two large-scale public remote sensing object detection datasets. Experiment results suggest that the proposed method is effective to alleviate the misalignment problem. Compared with the baseline model, the detection accuracy is improved by 8.5% mAP and 11.0% mAP on the challenging benchmark for object detection in optical remote sensing image (DIOR) and a large-scale dataset for object detection in aerial image (DOTA) dataset, respectively. Our best-resulting model achieves the state-of-the-art performance, surpassing other one-stage detectors both on the two datasets at a high detection speed of 21 FPS.**

***Index Terms*— Feature misalignment, object detection, one stage, remote sensing imagery.**

I. Introduction

With the rapid development of remote sensing technology, the interpretation of remote sensing imagery has become a popular field due to significant value both in academic study and engineering application. Object detection in aerial imagery is a fundamental subject in this field and has been widely explored and studied. To solve the problems shown in practical applications, many object detection algorithms [1]–[13] have been proposed and developed.

In recent years, deep learning-based object detection algorithms have been widely and continuously studied [5], [14]–[21]. With the development of high-resolution remote sensing technology, aerial images have larger sizes as well; therefore, high computation efficiency is required. Many one-stage detectors [22]–[27] have been proposed to achieve real-time detection. The majority of these detectors are based on the framework of the single-shot multibox detector (SSD) [23], which has a decent balance between speed and accuracy. Unfortunately, the efficient SSD shows an unsatisfactory accuracy when implemented into the remote sensing scene [28], [29]. Based on our study, we find that such detectors suffer from the feature misalignment problem. We analyze this problem from the following two aspects.

1) Misaligned Anchor: Fig. 1 shows the misalignment between the remote sensing object "Bridge/Ship" and its corresponding anchor, such positive anchors (yellow box) with low quality are called "*the misaligned anchor*" in this work. The characteristics of the objects in remote sensing scenes include multiple orientations, multiple aspect ratios, and multiple scales; thus, the anchor setting and matching are challenging, and the misaligned anchor is hence inevitable. In general, anchors bind objects and feature together by matching, so the misaligned anchor will lead to inadequate feature learning, which significantly limits the performance. Besides, the misaligned anchor will also damage the model convergence and lead to low regression precision.

2) Misaligned Feature: As shown in Fig. 1, the feature sampling points of the standard convolution kernel and the long-narrow geospatial object exhibit significant spatial misalignment; the collected feature is hence called "*the misaligned feature*" in this article. It is caused by the contradiction between the fixed convolution kernel and the variation of object shape. Although the theoretical receptive field is large

Manuscript received February 27, 2021; revised May 7, 2021 and June 7, 2021; accepted June 8, 2021. Date of publication June 29, 2021; date of current version January 12, 2022. This work was supported by the National Natural Science Foundation of China under Grant 61725105 and Grant 41701508. *(Corresponding author: Xian Sun.)*

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Digital Object Identifier 10.1109/TGRS.2021.3089170

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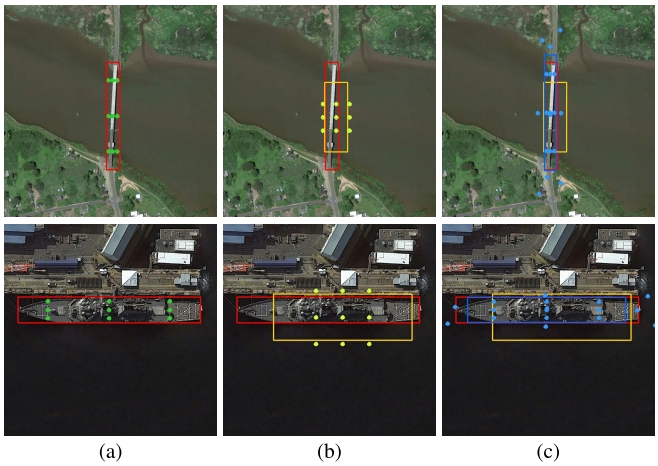


Fig. 1. Illustration of the misalignment problem and our aligned result. (a) Illustration of completely aligned convolution kernel (green points) and object (red rectangle). (b) In previous detectors, the predefined anchor (yellow rectangle), object bounding box (red rectangle), and the sampling points of the standard convolution (yellow points) are misaligned in spatial. (c) In our detector, the feature sampling points (blue points) are aligned with the object (red rectangle) in spatial according to the aligned anchor (blue rectangle).

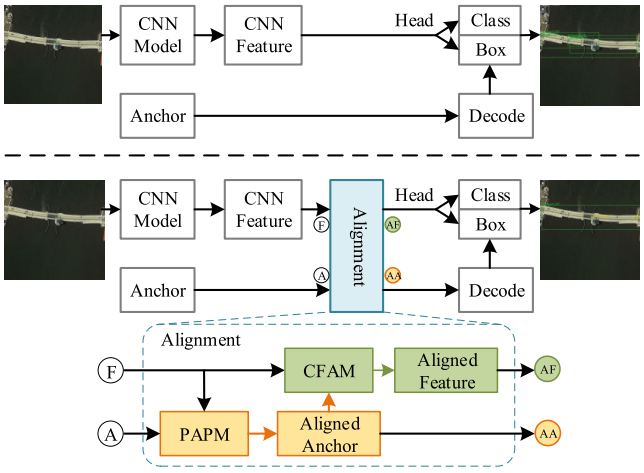


Fig. 2. Detection pipeline of previous detectors (top) and our proposed ASSD (bottom). Previous detectors directly regress and classify objects on the convolutional feature. Our detector leverages the proposed alignment modules to rectify the misalignment and then conducts detection on the aligned feature and aligned anchors.

enough, the effective receptive field is much smaller [30] and more misaligned, so the extracted convolutional features cannot completely cover the object and even contain noisy information. The spatial misalignment between features and objects will lead to poor consistency of features, resulting in low robustness and accuracy of the detector for remote sensing objects with multiple shapes and aspect ratios.

To address the misalignment problem introduced above, this article proposes a feature aligned single-shot detector (ASSD). Compared with the previous method as shown in Fig. 2, this detector contains two key alignment components: a pseudo anchor proposal module (PAPM) and a context-based feature alignment module (CFAM).

For the misaligned anchor, setting as many anchors as possible can guarantee high-quality matches. However, at the same time, it will bring redundant computation costs and aggravate the imbalance of positive and negative samples. To avoid this problem, we propose the PAPM to align the predefined anchors with potential objects. In this module, the predefined anchor group is replaced by a single square anchor, termed the "*core anchor*," to reduce redundancy and alleviate the severe negative/positive imbalance problem. The PAPM leverages a convolution subnet to refine the core anchor to provide the spatially aligned location initiation for the detection network.

For the misaligned feature, the essential solution is to deal with the contradiction between the fixed convolution kernel and the object shape variation. Based on this insight, we propose the CFAM to tackle this problem. The CFAM can adaptively adjust the sampling points of convolutional kernels according to the aligned core anchors and then collect aligned and consistent features. Besides, we append a group of additional sampling points to this module by expanding the core anchors so as to collect the contextual information; it can avoid the inappropriate sampling caused by the coarsely regressed core anchors. With the CFAM, the receptive field of the convolution kernel can adaptively transform into rectangles of arbitrary ratio instead of the fixed square so that the collected features can be adaptively aligned with objects.

The main contributions of this article are as follows.

1) We propose a feature aligned single-stage object detector to address the feature misalignment problem of the one-stage detector in the remote sensing scene; this detector balances the detection speed against accuracy and achieves the state-of-the-art performance on the remote sensing object detection datasets, benchmark for object detection in optical remote sensing image (DIOR) and a large-scale dataset for object detection in aerial image (DOTA).
2) For the misaligned anchor, a novel PAPM is proposed. This design can provide the anchor with a more aligned initial location and deal with the negative/positive imbalance.
3) For the misaligned feature, a flexible CFAM is proposed. This plug-and-play module can effectively rectify the misalignment between the sampling points of convolution kernel and objects, hence improving the consistency of features.

To validate the effectiveness of the proposed method, a series of comprehensive experiments are conducted on the large-scale public remote sensing object detection datasets, DIOR [29] and DOTA [28]. We equip the popular single SSD [23] with the proposed alignment components and then set up the ASSD. The experiment results show that the proposed method can effectively alleviate the misalignment problem and improves the detection accuracy by 8.5% and 11.0% mean average precision (mAP) on the challenging DIOR [29] and DOTA [28] dataset, respectively. The best-resulting model achieves 71.1% and 77.77% mAP on the task of detection with horizontal bounding boxes in these two datasets separately at a high detection speed of 21 frames/s (FPS), and it surpasses other state-of-the-art detectors on both datasets.

II. Related Works

The misalignment among the predefined anchor, object, and square convolution feature has caught the attention of

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researchers. Some algorithms hence have been proposed to address this issue from different aspects. Based on the research aspect, these methods can be roughly summarized as the anchor proposal mechanism-based methods and the feature alignment mechanism-based methods.

A. Anchor Proposal Mechanism

Some works start to think about the difference of detections with the anchor mechanism between the one- and two-stage detectors to build up an efficient detector with better accuracy. As discussed in the study of focal loss [27], the vast easy and negative anchors from the dense prediction in one-stage detectors can overwhelm training and lead to degenerate models. The focal loss is thus proposed to address this problem. Different from this strategy, the Faster R-convolutional neural network (CNN) family [4], [20] uses the region proposal mechanism to early filter out these invalid anchors; hence the further predictions only focus on the potential positive samples. Analogous to this mechanism, the one-stage RefineDet [31] and its variants [32]–[34] mimic the region proposal network (RPN) [16] to predict the score and refined location of anchors and then filter out the negative anchors. In the following step, further predictions are based on the refined anchors. The anchor proposal module (APM) in [35] is also designed to select positive anchors and get better initial locations and shapes. The guided anchor (GA) [36] can generate high-quality anchor proposals and filter out the negatives. These methods imitate the proposal mechanism [16] and cascade prediction [18] to eliminate the vast negative anchors and improve the regression precision. Inspired by this mechanism, we also leverage an additional subnetwork to predict the refined anchor. Unlike these methods, we only use one square anchor instead of the anchor group at each position on the feature map to early avoid redundant negative anchors. This design can inherently address this problem and fit the efficient fully convolutional network (FCN) architecture to speed up the anchor proposal procedure.

B. Feature Alignment Mechanism

Recently, some methods [32]–[39] have been proposed to tackle the spatial feature misalignment problem introduced above. The deformable convolution network (DeformConv) [37] can adaptively determine receptive field sizes by learning and adjusting the sampling locations in the convolution kernel. Thus, the deformable convolutional kernel can align the deformable receptive field with objects in spatial [37]. Inspired by this work, some object detectors leverage the DeformConv to align the receptive field with objects or anchors. We group these detectors into three different alignment families, as shown in Fig. 3(b)–(d). The standard convolution network without any alignment is shown in Fig. 3(a).

The method based on implicit alignment with DeformConv, such as the dual refinement network (DRNet) [32], RepPoints [38], and Cas-RetinaNet [39], follows the original deform pipeline in [37], as shown in Fig. 3(b). These models directly predict the offsets of DeformConv by taking the convolutional feature as input so that the feature sampling location is implicitly aligned to objects of varied shapes.

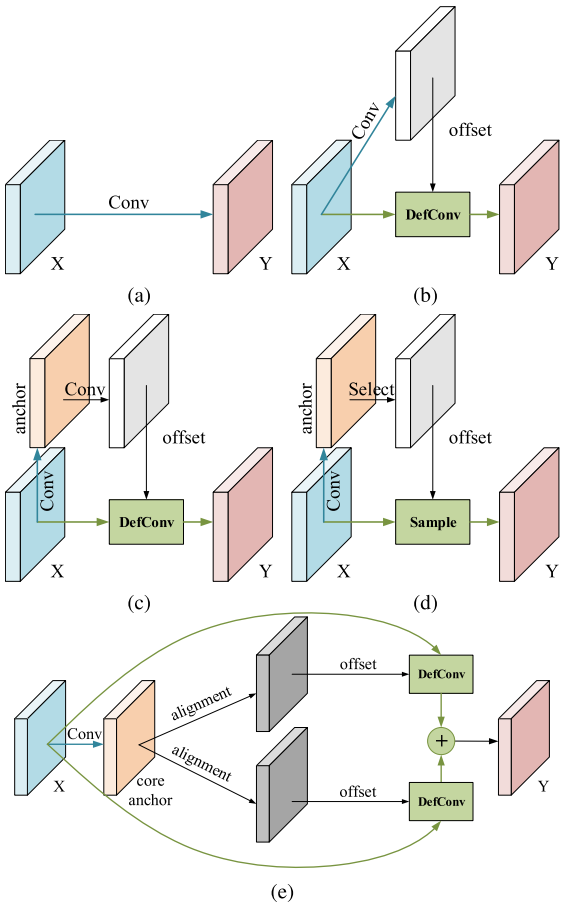


Fig. 3. Different architectures of the alignment methods. (a) Standard convolution network without alignment. (b) Implicit alignment with DeformConv (DefConv for short). (c) Alignment with anchor shape and location information. (d) Alignment with calculated sampling points. (e) Our alignment architecture.

This feature alignment strategy can adaptively deal with the misalignment problem by learning. However, this process is poorly interpretable, and the implicit alignment is too implicit to ensure that features are truly aligned in spatial [34].

The method of alignment with anchors, such as the study of [35] and GA-RPN [36], is shown in Fig. 3(c), and the sampling location offsets of DeformConv are predicted on the anchor shape and location information. The GA-RPN [36] is a representative study that is aimed to align the location and scale of the receptive field with anchors, in which the feature adaption learns offsets from the refined anchor shape by a regular convolution. Even though the feature alignment module (FAM) in [35], [36] can explicitly collect the shape and location information of the refined anchor to guide the DeformConv, the learned spatial alignment is still too implicit.

Fig. 3(d) shows the recent strategy to explicitly align the receptive field with objects by calculating sampling location instead of predicting. The study of R^3Det [34] selects one anchor at each feature point and then adaptively collects feature vectors at the center point and four corners of the refined rotated anchors, which can explicitly align the receptive field with the rotated geospatial objects. This method can tackle the spatial misalignment problem, but the cost is the long latency.

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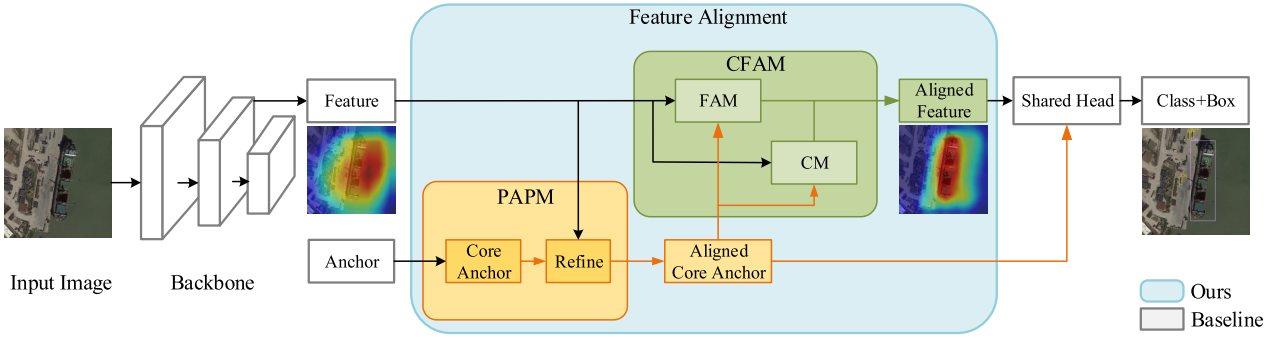


Fig. 4. There are four steps in this framework. First, the convolutional backbone network extracts the feature pyramid from the input image, of which each pyramid level would be fed into the following layers. Then, the PAPM regresses a group of localization offsets to align the core anchor with objects. Third, the CFAM leverages the aligned core anchor to guide the convolutional kernel to sample spatially aligned points to align features in spatial, and the outputs compose the aligned feature pyramid. Finally, the detection head is shared by all the aligned pyramid layers, which predicts the bounding box and category for each core anchor and produces the final results.

Inspired by these methods, we proposed our alignment strategy, as shown in Fig. 3(e). This architecture leverages the core anchor to provide spatial information and avoid computation for redundant anchors, and the two explicit alignment branches can collect aligned local and contextual features. The structures are detailed in Section III.

III. Methodology

In this work, we propose the ASSD to address the misalignment problem in one-stage detectors to detect multiscale objects in large-size remote sensing imagery. The overall architecture is shown in Fig. 4. This detector is based on the efficient single-stage framework and combined with the "anchor proposal" mechanism and the "adaptive feature alignment" mechanism to achieve efficient detection and high accuracy.

The ASSD is composed of the base framework and the proposed detection subnetwork. The main components in the subnetwork can be elaborated as follows: 1) PAPM: providing spatial information of the aligned core anchor and mitigating the neg/pos imbalance and 2) CFAM: aligning the feature sampling points of convolution kernel with objects in spatial.

A. Framework

The ASSD is based on the single-stage detection framework, which can efficiently detect geospatial objects with multiple scales. The Visual Geometry Group-16 (VGG-16) [40] is implemented to serve as the backbone network to extract features from aerial imagery. To build up the ASSD in FCN [41] style, the last classification layer of the backbone network is removed [23]. After that, a group of additional feature maps are appended to the last backbone layer to build up the feature pyramid, as shown in Fig. 4. Each pyramid layer deals with the objects with the corresponding scale. The anchor mechanism is implemented in the detection subnetwork to assign objects to reasonable layers and reduce the search space of prediction.

At each cell of the feature map in each pyramid level, a group of predefined anchors are refined by the predicted localization offsets. Unlike the prior works [34], there is only one core anchor in each anchor group because of the PAPM. After that, the refined anchors are fed into the CFAM to provide the spatial alignment information. Then, the bounding box predictions are conducted on the aligned feature maps. Finally, nonmaximum suppression (NMS) is adopted to reject redundant boxes and select decent results.

This framework is efficient and has a competitive accuracy when compared with two-stage advanced detectors. Thus, it is suitable for detecting multiscale objects in a large-size remote sensing image, which is time-consuming and challenging.

B. Pseudo Anchor Proposal Module

The widely adopted anchor is used to encode and decode the bounding box for regression, which can discretize and limit the solution space to ensure training stability. However, redundant anchors cause long latency and negative/positive imbalance. This problem makes the object detection of aerial imagery more challenging due to the large image size and sparse objects. The nature scene images are captured from the front view, which have severe overlap and ambiguity for anchor matching. For that reason, the dense anchors within a group, which have different aspect ratios and scales, can respond to different objects. While both aerial and satellite images are orthophoto with a top view, the overlaps among the geospatial objects are so small that the overlap and ambiguity problem can be ignored. To be specific, all the anchors within the group respond to one object at most, so the current anchor setting is redundant in this scene.

Based on this insight, the PAPM is proposed. As shown in Fig. 5(a), the PAPM inherits the strategy of anchor refinement module (ARM) [31], [34] to refine the anchor and make proposals. To prune the redundant anchors early, the PAPM uses a single anchor instead of a group of anchors for regression at each cell on the feature map. Note that the classification branch of the anchor group is removed because the anchor selection is useless for one anchor, which is quite different from other methods [31], [34], [42]. This design hence can be called the pseudo-APM. The usage of the anchor is split into two stages: matching and inference stage, as described in Algorithm.1.

1) Matching Stage: To ensure that all the objects are appropriately matched in the training process, the PAPM uses the same anchor group setting as normal [23]. Specifically, the anchor group at each cell on the feature map contains a

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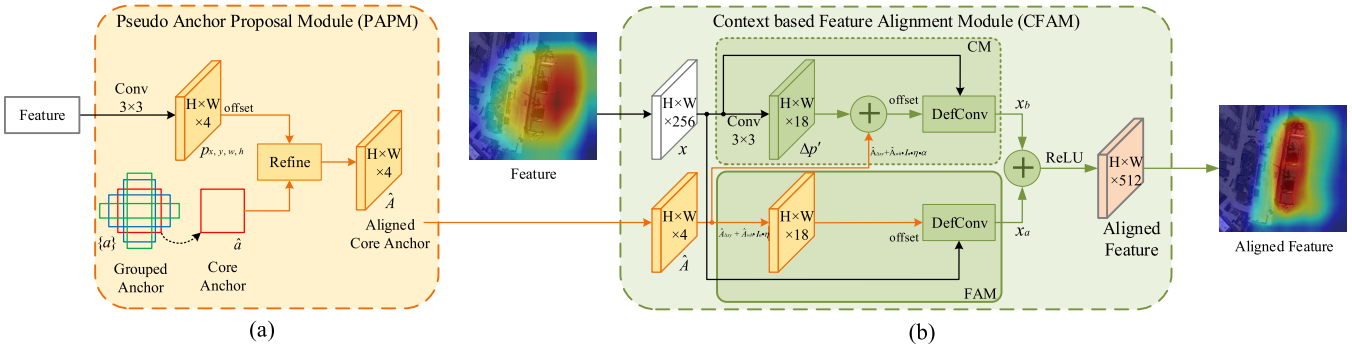


Fig. 5. Structure of the alignment modules. (a) Structure of PAPM. The grouped anchor is the regular anchor in previous detectors, and the core anchor means that only the base square anchor at each cell is remained to align and predict objects. (b) Structure of CFAM. Based on the aligned core anchor, the convolution kernel's sampling points are acquired and converted into offsets; then, the offsets are fed into the DeformConv to reconstruct aligned features. Then, the sampled region is extended to capture surrounding contextual features; besides, a group of offsets are predicted to adaptively adjust the contextual sampling region to prevent the low-quality core anchor.

Algorithm 1 PAPM, the Matching and Inference Stage Are Listed Before and After Line 10

Input: $\{g\}$ and $\{a\}$ are the ground-truth and predefined anchor group, Reg and x are the regression head and feature in PAPM, $\widehat{a}$ is the core anchor, θ is the matching threshold.

Output: L_r is the smooth L2 loss, $\widehat{A}$ is the set of aligned core anchors in PAPM.

1: Initialize $G \in \{\mathcal{R}^{H\times W\times 4}\}$, $\widehat{A} \in \{\mathcal{R}^{H\times W\times 4}\}$, $L_r \in \{\mathcal{R}^{H\times W}\}$ for L feature layers.
2: $\{G_m\}, \{S_m\} \leftarrow \text{IOUMATCH}\{g\}, \{a\}, \theta$
3: # $\{G_m\}$ is matched ground-truth, $\{S_m\}$ is matched IoU.
4: **for** each $l \in [1, L]$ **do**
5: **for** each $i \in [1, H]$ **do**
6: **for** each $j \in [1, W]$ **do**
7: $id = \text{ARGMAX}S_m[l, i, j]$
8: $G[l, i, j] = G_m[l, i, j][id]$
9: $t_{x,y,w,h} = \text{ENCODE}G[l, i, j], \widehat{a}[l, i, j]$
10:
11: $p_{x,y,w,h} = \text{REG}x_l[i, j]$
12: $L_r[l, i, j] = \text{SMOOTH_L2}p_{x,y,w,h}, t_{x,y,w,h}$
13: $\widehat{A}[l, i, j] = \text{DECODE}p_{x,y,w,h}, \widehat{a}[l, i, j]$
14: **end for**
15: **end for**
16: **end for**
17: **return** $L_r, \widehat{A}$

series of anchors with varied aspect ratios and scales. The ground-truth bounding boxes $\{g\}$ are assigned to corresponding anchors $\{a\}$ if the intersection-over-union (IoU) is higher than the threshold θ. After the preliminary matching, each predefined anchor has a matched ground truth g_m. Then, we compare the matching score s_m (the IoU matrix) within each anchor group separately. For each feature cell, only the one ground truth with the maximum matching score is remained to be the final target. In this way, many matched objects will be dropped out due to the low matching score, which means that the low-quality samples are filtered out; hence, the training process is more stable. It worth noting that the anchor group $\{a\}$ is used to assign ground truth to the core anchor only in training, which is a crucial strategy of the PAPM.

2) Inference Stage: In the inference stage, only one single anchor in each group is used to refine. Unlike the matching stage, each anchor group in this stage contains only one square anchor, which is called the "*core anchor*" $\widehat{a}$. Compared to the normal designs [31], [34]–[36], this module merges the search space of prediction within each anchor group to a new space. It can effectively reduce the computation and latency. To get the aligned anchor, the PAPM predicts the coordinate offsets $p_{x,y,w,h}$ to refine each core anchor $\widehat{a}$ by the following decoding equations:

$$\widehat{A}_x = p_x \cdot \widehat{a}_w + \widehat{a}_x, \widehat{A}_y = p_y \cdot \widehat{a}_h + \widehat{a}_h \tag{1}$$

$$\widehat{A}_w = e^{p_w} \cdot \widehat{a}_w, \widehat{A}_h = e^{p_h} \cdot \widehat{a}_h \tag{2}$$

where $\widehat{A}$ is the aligned core anchor. The target of $p_{x,y,w,h}$ is acquired by the inverse operation of 1 and 2, namely the encoding step of Algorithm 1 (line 9).

3) Setting of Anchor Scale: To get a better scale assignment for the core anchor of PAPM, the anchor generator is reformulated in this section. A popular anchor scale assignment is the linear scale arrangement [23]. This strategy can regularly discretize scales to each layer, but it ignores the issue of the mismatched receptive field. The exponential scale arrangement [27], [34] enlarges the anchor scales layer by layer in an exponential manner, which can match the scale increment with the feature steps. However, there are more than five detection layers in ASSD so that the anchor scales on the deeper layers vastly exceed the imagery size. Hence, the anchors in the last two layers will not match any objects.

To this end, we design a more flexible anchor generator to assign a series of scales to the anchors layer by layer. This generator arranges the anchor scales based on the feature scale and steps to deal with the scale mismatch problem; it is hence called the scale aligned anchor generator (SAA). Specifically, it is formulated as follows:

$$s_k = \begin{cases} s_0 \cdot 2^{k-1} & k \in [1, m] \\ s_0 \cdot 2^{m-1} + \dfrac{s_{\max} - s_0 \cdot 2^{m-1}}{L - m}(k - m) & k \in [m+1, L] \end{cases} \tag{3}$$

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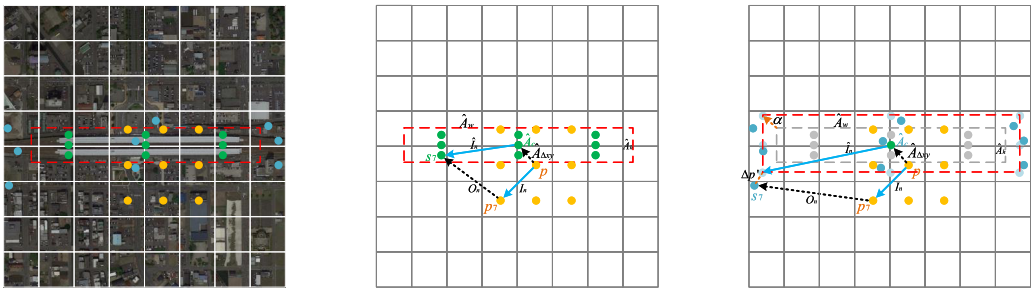


Fig. 6. Illustration of the sampling points of the CFAM. The red dashed box is the core anchor refined by the PAPM. The orange points show the standard convolution kernel's sampling position on the feature cell (white grid) p. The green and cyan points show the aligned sampling position of FAM and context module (CM), respectively. Better view in color. (a) CFAM. (b) FAM. (c) CM.

where s_0 and $s_{\max}$ are the lower and upper limits of the anchor scale, L is the number of detection layers, and m is a flexible parameter to adjust the scale arrangement according to the specific dataset, and we empirically set it to 3 to fit the DIOR and DOTA dataset in this article. When m is set to 1, 3 degenerates into the linear scale generator, and the exponential scale generator when $m = L$. This flexible scale arrangement unifies the linear and exponential scale arrangement to be easily adapted to other detectors.

In the matching stage of the PAPM, we set the base scale of each anchor group following the SAA to match objects of varied scales better. In the inference stage, we keep the base square anchor in each anchor group as the core anchor $\widehat{a}$. Thus, the objects matched by the core anchors are also aligned to the receptive fields in scale.

C. Context-Based Feature Alignment Module

The RoIAlign [21] utilizes feature sampling and pooling operations to reconstruct features so that the features collected by each region of interest (ROI) are spatially aligned with the objects. This design improves the performance of the detection subnetwork from two aspects: 1) it improves the feature consistency and hence can speed up the model convergence and 2) the flexible receptive field can adapt to objects of various shapes and scales, which is more conducive to collecting these objects' intact features. Inspired by this design, we propose the CFAM for the one-stage detector. This module utilizes the spatial feature reconstruction strategy such as the RoIAlign to solve the spatial misalignment; besides, it can collect the contextual information to acquire more discriminative features.

1) Feature Alignment Module: As shown in Fig. 5(b) and Fig. 6(b), the FAM leverages the aligned core anchor $\widehat{A}$ to get the spatial information of corresponding objects so as to guide the feature alignment. Specifically, the PAPM can coarsely regress the bounding box of each object based on the core anchor. Then, the region in each aligned anchor is taken as the potential region where the object lies. To collect intact features in the target region, we regularly sample $k \times k$ feature points in each aligned core anchor box $\widehat{A}$. More specifically, we use the location and shape of core anchor $\widehat{A}$ to recalculate the offsets of the DeformConv kernel at each feature cell separately. It is quite different from the standard convolution and the deformable convolution [37].

For the standard convolution, the feature vector of cell p on the output feature map x_a is collected from the input feature map x by a $k \times k$ square kernel w

$$x_a(p) = \sum_{n=1}^{k\times k} w(I_n) \cdot x(p + I_n) \tag{4}$$

where I_n is the inner offsets within the kernel w centered on the cell p, taking the example of a 3×3 kernel, I_n is a group of coordinates offsets

$$I_n = \{(-1,-1),(0,-1),(1,-1)\ldots,(0,1),(1,1)\}. \tag{5}$$

For the deformable convolution, the kernel is not a regular square anymore. The sampling positions of the deformed kernel are decided by the predicted offsets O_n, and then, 4 evolves into the following formula:

$$x_a(p) = \sum_{n=1}^{k\times k} w(I_n) \cdot x(p + I_n + O_n) \tag{6}$$

where the O_n is predicted by an convolutional subnetwork in the DeformConv [37], as shown in Fig. 3(b).

For the FAM, the offsets are calculated by the refined core anchor $\widehat{A}$ instead of the implicit convolutional prediction. As shown in Fig. 6(b), the inner sampling point s_n can be represented by vectors; then, we have

$$\begin{aligned} \overrightarrow{pp_n} + \overrightarrow{p_n s_n} &= \overrightarrow{p\widehat{A}_c} + \overrightarrow{\widehat{A}_c s_n} \\ I_n + O_n &= \widehat{A}_{\Delta xy} + \hat{I}_n \\ &= \widehat{A}_{\Delta xy} + \widehat{A}_{wh} \cdot I_n \cdot \eta \end{aligned} \tag{7}$$

where η is a hyperparameter to control the sampling position within each region. We regularly divide the region centered on $\widehat{A}_c$ into $k \times k$ grids and sample features at the center of each grid, so η is set to $1/k$. $\widehat{A}_{\Delta xy}$ and $\widehat{A}_{wh}$ are the location offsets and shape of the refined core anchor, respectively, which provide the spatial information of objects to guide the feature reconstruction and alignment.

Unlike the RoIAlign, we sample feature points by the bilinear interpolation without pooling and cropping to keep the efficiency. In this way, the feature collected by the FAM is explicitly aligned to the potential object region. Compared to

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the standard convolution, this flexible sampling kernel is more adaptive to the geospatial objects of various shapes, such as the long-narrow ship, bridge, and harbor, which are difficult for the regular fixed square convolution to capture intact and aligned feature.

2) Context Module: Nevertheless, the collected feature still can be misaligned with the corresponding object because the refined core anchor can not wholly coincide with the ground-truth bounding box. Thus, the coarsely regressed anchor may mislead the kernel to sample inappropriate points. To this end, we add a group of more flexible sampling points to this module, as shown in Fig. 6(c). The additional feature points are acquired by expanding the sampling region in FAM by α; these points are called the outer points. Then, we add a group of trainable implicit offsets $\Delta p'$ to each outer point to adjust the outer region adaptively. This design can collect contextual information and rectify the inaccurate sampling positions. This module can be described as follows:

$$x_b(p) = \sum_{n=1}^{k\times k} w(I_n) \cdot x(p + \widehat{A}_{\Delta xy} + \widehat{A}_{wh} \cdot I_n \cdot \eta \cdot \alpha + \Delta p') \tag{8}$$

where α is the region expanding factor and it can expand the kernel like the dilated convolution [43] and $\Delta p'$ is the additional offset prediction like O_n in 6.

Finally, we merge the local and contextual feature to reconstruct the aligned feature maps, as shown in Fig. 5(b); this process can be described as

$$X(p) = \mathrm{ReLU}(x_a(p) + x_b(p)). \tag{9}$$

D. Detection Head

The detection head of ASSD is composed of the standard convolution. The CFAM has collected the aligned feature for each feature cell, so the detection head can only adopt the 1×1 convolutional kernel to conduct the classification and regression. Inspired by the detection head in two-stage detectors [16], we use two stacked 1×1 convolution to imitate the shared multilayer perceptrons (MLP). Note that all the detection layers share this head to prevent redundant parameters. Besides, the shared head can alleviate the lack of positive samples caused by the sparse geospatial objects and dense anchors and improve the convergence speed.

1) Inference: The detection head predicts a $C + 1$ score vector and a four-tuple (x, y, w, h) for classification and regression at each cell. Note that the head predicts only one bounding box for the aligned core anchor instead of a group of boxes, which is significantly different from other anchor-based detectors. Then, the class scores are normalized by the softmax layer, and the predicted bounding box B is decoded by the refined core anchor $\widehat{A}$ as follows:

$$B_x = x \cdot \widehat{A}_w + \widehat{A}_x, B_y = y \cdot \widehat{A}_h + \widehat{A}_y \tag{10}$$

$$B_w = e^w \cdot \widehat{A}_w, B_h = e^h \cdot \widehat{A}_h. \tag{11}$$

Finally, all the predicted bounding boxes are collected and processed by NMS to reject redundant boxes and select the possible optimal results.

E. Loss Function

The overall objective function of the ASSD is composed of the loss from the PAPM and the detection head. In the PAPM, each feature cell merely predicts the location and shape of the core anchor, so only the regression loss of positive samples is considered. This loss is formulated as follows:

$$L_r = \frac{1}{N_{\mathrm{pos}}} \sum_{i\in \mathrm{pos}} \sum_{m\in\{x,y,w,h\}} \mathrm{smooth}_{L1}(p_i^m, t_i^m) \tag{12}$$

where p^m and t^m denotes the regressed coordinates $\{x, y, w, h\}$ and its targets, respectively, i is the index of positive core anchors, N_{pos} is the number of positive anchors in the training stage, and the function of smooth_{L1} is the smooth L1 loss which is defined as

$$\mathrm{smooth}_{L1}(x) = \begin{cases} 0.5x^2, & if \quad |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases}. \tag{13}$$

In the detection head, the loss contains the regression loss and classification loss as usual. The regression loss L_{reg} has the same formula as 12, and it worth noting that the targets are matched and encoded by the refined core anchor instead of the standard anchor groups; the encoding formula is the inverse operation of 10 and 11. The classification loss is the softmax loss over multiple classes confidences

$$L_{\mathrm{conf}} = \frac{1}{N_{\mathrm{pos}}} \left(-\sum_{i\in \mathrm{pos}} y_i \log(y_i, p_i) - \sum_{i\in \mathrm{neg}} \log(p_i) \right) \tag{14}$$

where N_{pos} denotes the total number of positive aligned core anchors and y_i and p_i are the class label and predicted posterior distribution over classes. We choose the positive core anchors by matching the ground-truth bounding boxes with the refined core anchor $\widehat{A}$ from the PAPM and the negative anchors by the hard negative mining strategy [23].

Finally, the overall objective function is the sum of the loss functions introduced above

$$L = L_r + L_{\mathrm{reg}} + L_{\mathrm{conf}}. \tag{15}$$

IV. Experiments

This section details the experimental scheme, including the description of datasets, evaluation metrics, implementation details, comparative experiments, analysis of results, and our discussion.

A. Datasets

Remote sensing objects have multiple scales and extreme aspect ratios, so misalignment in target detection is severe. Therefore, we conducted experiments on two publicly available large-scale remote sensing object detection datasets, DOTA and DIOR.

1) DIOR: We evaluate the effectiveness of our method on the optical remote sensing dataset DIOR [29]. The DIOR dataset contains 23 463 images with a fixed size of 800×800 pixels, in which 192 472 instances are labeled by horizontal bounding boxes and categorized as 20 geospatial object classes shown in Fig. 7. This dataset has a wide range of object

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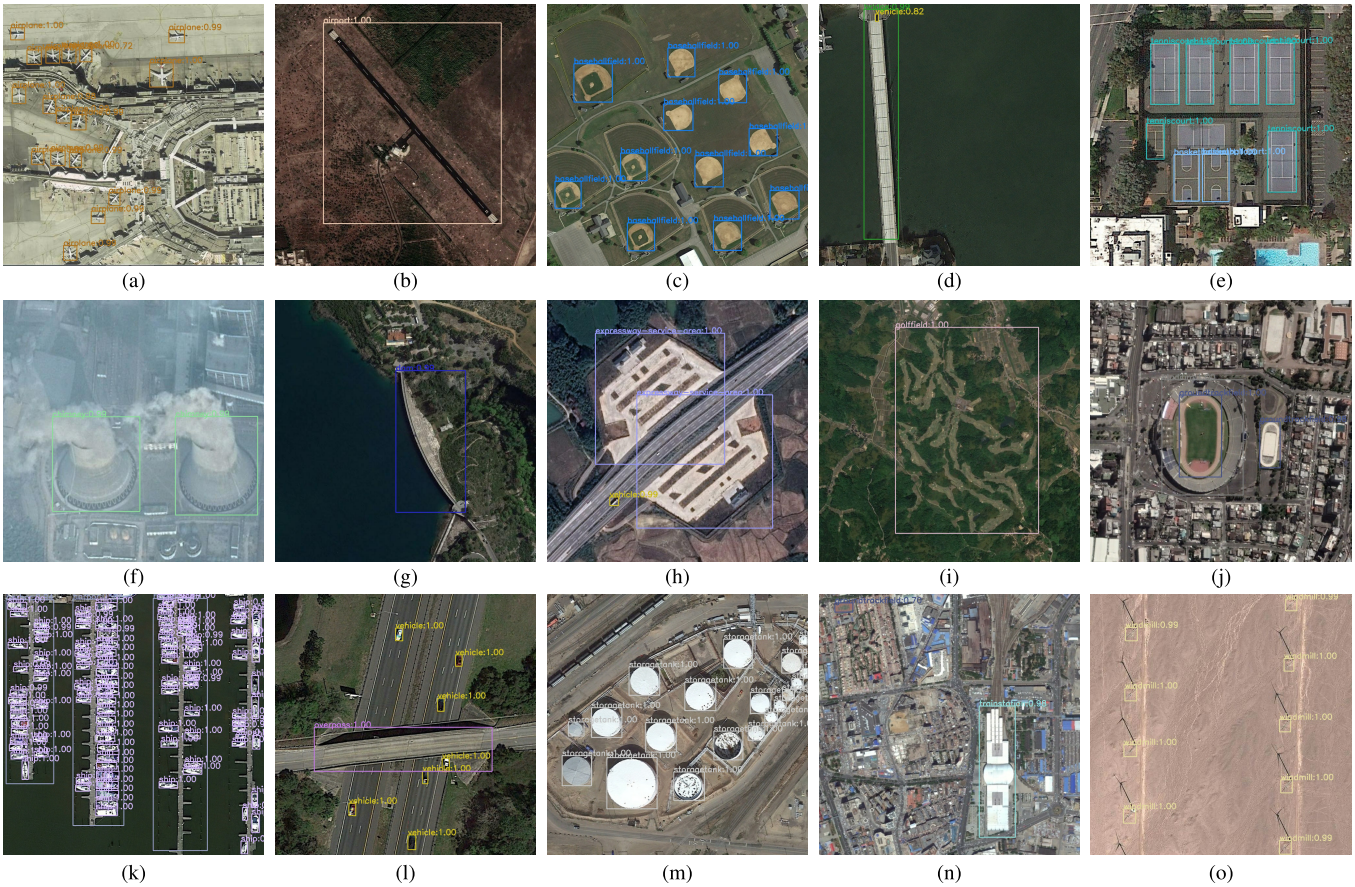


Fig. 7. Detection results of ASSD for each category in the DIOR dataset. (a) Airplane. (b) Airport. (c) Baseballfield. (d) Bridge. (e) Basketballcourt and Tenniscourt. (f) Chimney. (g) Dam. (h) ESA. (i) Golffield. (j) Ground-track-field and Stadium. (k) Harbor and Ship. (l) OP and Vehicle. (m) Storagetank. (n) Trainstation. (o) WM.

and image variations, such as the spatial resolution, the imaging conditions, and the scales of objects; hence, it is quite challenging [29]. The DIOR is officially divided into three parts: the ratio of the train set, val set, and test set is $1:1:2$. For the simplicity of expression in experimental results, the 20 categories (c1–c20) are written as follows for short: airplane (APla), airport (APort), baseball field (BF), basketball court (BC), bridge (Bri), chimney (Chi), dam (Dam), expressway-service-area (ESA), expressway-toll-station (ETS), golf field (GF), ground track field (GTF), harbor (Har), overpass (OP), ship, stadium (Stad), storage tank (ST), tennis court (TC), train station (TS), vehicle (Veh), and windmill (WM).

2) DOTA: To inspect the generalization of our method, the experiments are also conducted on the DOTA [28] dataset. The DOTA dataset collects 2806 high-resolution aerial optical remote sensing images, of which the sizes range from 800×800 to 4000×4000 pixels, as shown in Fig. 8. Totally, 188 282 object instances are annotated by arbitrary quadrilateral bounding boxes. These instances are categorized as 15 common object categories: plane (Pla), baseball diamond (BD), bridge (Bri), ground field track (GTF), small vehicle (SV), large vehicle (LV), ship, TC, BC, ST, soccer ball field (SBF), roundabout (RA), harbor (Har), swimming pool (SP), and helicopter (HC). Following the official data segmentation, the ratio of the train set, val set, and test set is $3:1:2$. Limited by the hardware resources, all of the 1411 raw training images are cropped into slices of 800×800 pixels with an overlap of 100 pixels. Finally, we get a total of 11 977 slices as the training set. For testing, the 937 original testing images are cropped into slices and resized into 512×512 pixels, and then, these chips are detected one by one. We collect all of the detection results from each slice by a global NMS, so as to compose the detection result of the intact original imagery. Finally, the evaluation metric is calculated on the 937 large-size original aerial images by submitting the results to the evaluation website [28].

B. Evaluation Metric

The widely adopted evaluation metrics for object detection are also introduced to evaluate the performance of our method. Specifically, the mAP is used to evaluate the overall detection accuracy, and the FPS is adopted to estimate the detector's inference speed. In addition, we evaluate the mAP under three different localization precision thresholds to show the performance of our proposed components, i.e., mAP_{50}, mAP_{75}, and mAP_{90}.

As discussed in Section III-C, the ASSD can adapt to objects of varied shapes and aspect ratios. To validate this point, we also evaluate the detection performance of the ASSD over geospatial objects with different aspect ratios and scales by the diagnosing tools [44]. This tool evaluates the normalized average precision (AP_N) of five groups of objects,

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Fig. 8. Detection results of ASSD for each category in the DOTA dataset.

of which each group has a unique range of aspect ratios/scales. Specifically, the group "T" and "XT" means the tall and extra-tall objects, "M" means the objects with small aspect ratios, "W" and "XW" means the wide and extra-wide objects, and the scales are grouped in the same way, i.e., "XS"/"XL," "S"/L," and "M." Note that "AP_N" is different from the "AP" metric; the formula of AP is shown as follows:

$$\mathrm{AP}(c) = \frac{R(c) \cdot N_j}{R(c) \cdot N_j + F(c)} \tag{16}$$

where $R(c)$ is the recall with confidence of at least c, N_j is the number of objects in class j, and $F(c)$ is the number of false positive detections with at least c confidence [44]. The number of remote sensing objects has a quite imbalance distribution; thus, the category with a large number would have a higher AP than if N_j is small for the same detection rate [44]. The normalized precision (AP_N) is designed to address this problem

$$\mathrm{AP}_N(c) = \frac{R(c) \cdot N}{R(c) \cdot N + F(c)} \tag{17}$$

where $N = 0.15 \cdot N_{images}$ is a category-diagnostic normalization factor, which can reduce the sensitivity of image number for specific class [44].

C. Parameter Settings and Implementation Details

The detector ASSD is built on the vanilla single SSD [23] in a fully convolutional framework to keep high efficiency. The backbone is initialized by the public pretrained models on the ImageNet [45]. The extra feature layers in the feature pyramid are initialized by the Xavier method [23].

In the ASSD, the scales of the core anchor are set following equation (3). The parameter m is decided by the object size distribution of the specific dataset. In the DIOR [29] and DOTA [28] dataset, small objects have a large proportion, so we set m to slow down the scale increment of the shallow layers to ensure that more small objects can be matched better. Our preliminary experiments show that the detector can be stable and achieve the best accuracy when $m = 3$. For anchor group $\{a\}$ of PAPM, the base scales are set in the same way, and the aspect ratios are set to $\{1, 1/2, 2, 1/3, 3\}$ except for the last two layers that are set to $\{1, 1/2, 2\}$. To match the anchor group with ground truth, we set the matching threshold to 0.5 for positive and 0.4 for negative. The aligned core anchor has a more precise location, so we set the matching threshold to 0.7 for positive and 0.5 for negative.

In the preprocessing of training, the data augmentation from SSD [23] is introduced. In the inference stage, the images in the DOTA dataset are detected piece by piece. Each image piece retains up to 300 bounding boxes after the local NMS under the overlap threshold of 0.5. Then, the retained boxes are aggregated again by a global NMS under the overlap threshold of 0.5 for each class to choose the final bounding boxes for each big image. The detection of DIOR is the global detection one by one without cropping.

The stochastic gradient descent (SGD) optimizer is adopted to train our detector with a mini-batch size of 16. The learning rate is set to 0.001; then, it is decreased by a factor of 0.1 after

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TABLE I

OVERALL PERFORMANCE OF THE ABLATION EXPERIMENTS ON THE DIOR DATASET. THE BASELINE MODEL IS OUR REIMPLEMENTED SSD DETECTOR ON THIS DATASET, THE LAST MODEL IS OUR PROPOSED ASSD DETECTOR

overall performance									size					aspect ratio					
#	Baseline	PAPM	FAM	CFAM	SAA	mAP_{50}	mAP_{75}	mAP_{90}	AP_{XS}	AP_S	AP_M	AP_L	AP_{XL}	AP_{XT}	AP_T	AP_M	AP_W	AP_{XW}	APort
1	√					62.57	41.77	9.52	9.60	31.95	64.95	85.90	90.40	60.20	64.70	61.45	56.00	53.90	69.91
2	√	√				64.56	42.55	10.01	10.60	34.65	66.80	86.70	90.85	62.30	66.35	62.70	58.25	55.95	72.43
3	√	√	√			67.20	46.71	11.35	11.15	38.45	72.65	90.80	91.40	66.85	70.65	65.50	63.25	62.70	73.76
4	√	√	√	√		67.75	47.63	12.68	12.50	39.55	73.35	90.85	90.85	67.30	71.00	66.20	63.85	63.60	71.84
5	√	√	√	√	√	**71.13**	**49.85**	**13.46**	**15.55**	**43.70**	**77.45**	**92.00**	**92.30**	**70.60**	**73.85**	**69.45**	**67.65**	**66.50**	**82.42**

extreme aspect ratio									normal aspect ratio										
#	Bri	Dam	GTF	Har	OP	Ship	TS	Veh	APla	BF	BC	Chi	ESA	ETS	GF	Stad	ST	TC	WM
1	30.35	55.35	69.16	57.04	53.66	62.09	54.05	26.84	72.96	70.96	86.83	**78.33**	67.12	49.50	78.00	75.10	40.11	85.17	68.94
2	31.89	58.02	70.89	57.27	55.02	65.56	55.03	30.49	76.75	72.80	87.58	77.81	**68.85**	52.37	80.04	**78.15**	42.89	86.29	71.06
3	33.77	58.05	71.98	59.65	56.33	78.04	61.29	37.99	79.70	74.46	87.22	76.79	66.97	55.73	79.04	75.30	57.41	85.35	75.21
4	34.72	58.45	71.99	60.52	56.58	79.05	62.28	38.59	80.85	74.91	86.81	78.00	68.33	56.11	78.56	76.71	58.19	85.89	**76.71**
5	**40.66**	**64.72**	**78.62**	**61.99**	**58.01**	**84.87**	**62.40**	**44.52**	**85.58**	**75.84**	**89.47**	77.60	67.10	**61.66**	**80.81**	76.71	**65.32**	**87.88**	76.34

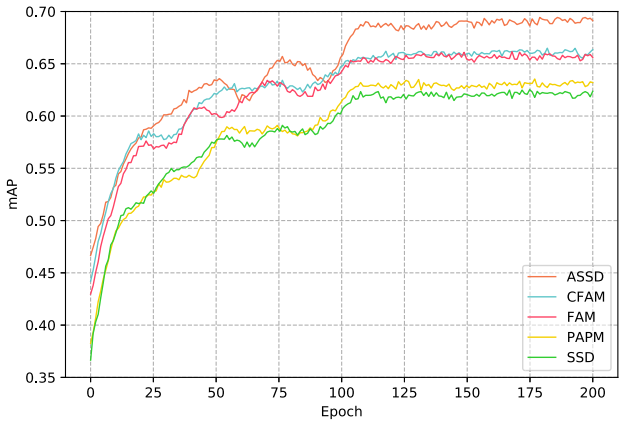


Fig. 9. Performance curves for the ablation experiments. We evaluate the performance of each training epoch on the validation set. The Y-axis shows the mAP_{50} of different methods as the training time increases.

100 and 150 epochs. In addition, we set the weight decay factor to 0.0005 and set the momentum to 0.9. To speed up the training process, we use an NVIDIA Tesla V100 GPU to train each model for 200 epochs. Finally, we evaluate the FPS on an NVIDIA GeForce 1080Ti GPU.

D. Quantitative Analysis

Following the scheme described above, we build up the ASSD model. A series of ablation experiments are then conducted on the DIOR and DOTA dataset to inspect and analyze the effectiveness of our method. Finally, we compare our method's performance with the state-of-the-art methods on the DIOR and DOTA datasets.

1) Ablation Experiments: The ablation experiments are conducted on both two datasets. The detection performance is quantitatively evaluated with the mAP metric for accuracy, FPS for speed, and AP_N for varied aspect ratios and scales. The mAP of the DOTA dataset is acquired by submitting the detection results to the DOTA evaluation website [28]. The performance on DIOR is evaluated on the publicly accessible testing set. The main quantitative results can be seen in Tables I and IV. The quantitative results show that our proposed components can progressively improve the overall accuracy. In Fig. 9, we compare the performance curves on the validation set in the training process, and the performance gap between the curves shows the effectiveness of each component.

TABLE II

ABLATION EXPERIMENT OF THE PAPM ON THE DIOR DATASET

PAPM	#total	#pos	#neg	$\frac{\#neg}{\#pos}$	mIoU	FPS	mAP_{50}
×	32736	31	32682	1055	0.74	30	62.57
√	5456	27	5429	**201**	**0.83**	**35**	**64.56**

a) Baseline setup: The baseline model is the single SSD [23], of which the backbone is the pretrained VGG16. The input size is set to 512 to ensure that the small remote sensing objects can be detected better so as to build up a robust and competitive baseline. On the DIOR dataset, the overall testing mAP_{50} is 62.57% at a speed of 30 FPS. Our reimplemented performance is close to the official result [29], which is even better than some two-stage detectors. On the DOTA dataset, this model achieves 64.99% mAP_{50}.

b) Effect of PAPM: To inspect the effectiveness of the PAPM, we compare the performance of the model with/without this module in Tables I and II. The results show that the PAPM improves the overall detection accuracy by 2.0 mAP_{50} and improves the inference speed by 5 FPS. Compared to the baseline model, the PAPM uses our proposed core anchor instead of the anchor group to predict potential boxes. Thus, the total number of anchors (denoted as "#total") becomes less than the baseline, and the detection becomes faster. In Table II, we report the averaged number of positive and negative anchors per image, denoted as "#pos" and "#neg," the comparison between them is quantitatively evaluated by the value of $\#neg/\#pos$. The results show that the PAPM can significantly alleviate the neg/pos imbalance. Also, we calculate the quality of the final anchor matching by the mean IoU (mIoU) scores of all matched positive anchors across all images in Table II. The improved mIoU shows that the PAPM can provide better spatial initialization than the predefined anchors by hand. In other words, the core anchors refined by the PAPM are spatially aligned to the object bounding boxes better. The comparison between aligned core anchors and misaligned predefined anchors is shown in Fig. 10.

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Fig. 10. Comparison between the predefined anchors and the aligned core anchors. The predefined anchors (yellow rectangle) are misaligned to the objects (red rectangle). After the PAPM, the core anchors (blue rectangle) are aligned with objects better, and the positive core anchors are much more than the positive predefined anchors.

c) Effect of FAM: The proposed FAM in ASSD is designed to address the feature misalignment issue of the feature extraction in the fully convolutional one-stage detector. As shown in Table I, the FAM can boost the overall accuracy by 2.64 mAP_{50} and 4.16 mAP_{75}, respectively, which shows the effectiveness of this module. As analyzed before, the fixed square convolution kernel of the baseline model is not appropriate to extract the feature of geospatial objects with varied shapes because of the spatial misalignment, while our proposed FAM can ease this problem. To validate this point, we evaluate AP_N of five groups of objects with different aspect ratios by the diagnosing tools [44], and the quantitative results are reported in Table I and Fig. 11.

The evaluation results show that the baseline method has worse accuracy for the wide (56.0%) and extra-tall/wide (53.9%) objects than the middle objects (61.45%). It is because the accumulative spatial misalignment in the convolutional layers makes it difficult for the vanilla kernel to capture the long-narrow objects. It hence can be found that the performances of long-narrow "Harbor," "Overpass," and "Bridge" in Table I are poor. After adding the FAM, AP_N improvement becomes higher and higher as the aspect ratio increases in Fig. 11(c) (pink curve on the right), which means that the FAM is more useful for the long-narrow objects ("XT," "T," "W," and "XW") than the square-like middle objects ("M"). Besides, the AP_N improvement of the long-narrow "Harbor" and square "Storagetank" in Fig. 11(a) and (b) can also confirm this point. It is because the spatial misalignment of long-narrow objects is worse than others, while our proposed FAM can effectively rectify it.

d) Effect of CFAM: Considering that the poorly regressed core anchors would mislead the FAM to sample the wrong location, we design the CM to rectify the misaligned sample regions and adaptively collect surrounding contextual features in the FAM. To validate this point, we evaluate the mAP metric for the overall accuracy under different regression precision thresholds (0.50, 0.75, and 0.90). The results are listed in Table I, which shows the performance of the well and poorly regressed boxes. Besides, the AP_N for five groups of objects with different sizes ("S"/"L" means small/large size, "M" means middle size, and "X" means "extra-") are also evaluated in Table I because the contextual information is more effective for small objects than big objects. Finally, the AP_N improvement curves are sketched in the left of Fig. 11(c).

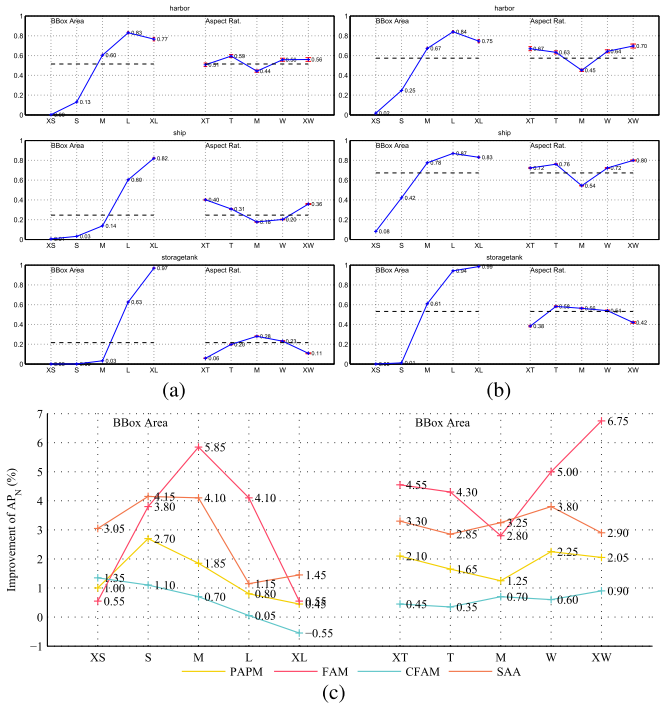


Fig. 11. Effect of the different method on varied scales/areas ("XS," "S," "M," "L," and "XL") and shapes/aspect ratio ("XT," "T," "M," "W," and "XW") on the DIOR dataset. (a) Baseline model's performance for the category of "Harbor," "Ship," and "Storagetank." Y-axis shows the value of AP_N metric of varied scales and shapes. (b) Results of the proposed ASSD. (c) Performance improvement of each module in the ablation experiment.

The results in Table I show that the FAM mainly improves the mAP under the low threshold (0.50 and 0.75). The core anchors refined by the PAPM are still not entirely aligned to the target region in spatial. Hence, the features collected by the FAM are still misaligned. Thus, the accuracy under the high threshold (0.90) is improved less than lower thresholds. On the contrary, the mAP improvement of the CFAM grows as the localization precision threshold. It validates our hypothesis that the CM can rectify the poorly regressed sampling and hence makes detection more precise. The pink curve in the left of Fig. 11(c) shows that the FAM mainly improves the objects with middle size instead of small/large and extra-small/large size. The reason is that the large-size objects are predicted in the deeper layers, which have larger receptive fields, and hence the impact of the misalignment problem and the FAM is small. For the small objects, the FAM would shrink the kernel to

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TABLE III

ACCURACY (%) COMPARISON OF STATE-OF-THE-ART METHODS ON THE DIOR DATASET. "SSD*" MEANS OUR REIMPLEMENTED BASELINE MODEL, "ASSD" IS OUR PROPOSED METHOD, "HOUR104" MEANS HOURGLASSNET104, AND "DARK53" MEANS CSPDARKNET-53. THE CLASS NAME C1-C20 CAN BE SEEN IN SECTION IV-A1

Methods	Backbone	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12	c13	c14	c15	c16	c17	c18	c19	c20	mAP	FPS
Two-stage methods																							
CornerNet [46]	Hour104	58.8	**84.2**	72.0	80.8	**46.4**	75.3	64.3	81.6	**76.3**	79.5	79.5	26.1	60.6	37.6	70.7	45.2	84.0	57.1	43.0	75.9	64.9	-
MRCNN [21]	ResNet101	53.9	76.6	63.2	80.9	40.2	72.5	60.4	76.3	62.5	76.0	75.9	46.5	57.4	71.8	68.3	53.7	81.0	62.3	43.0	81.0	65.2	11
PANet [47]	ResNet101	60.2	72.0	70.6	80.5	43.6	72.3	61.4	72.1	66.7	72.0	73.4	45.3	56.9	71.7	70.4	62.0	80.9	57.0	47.2	84.5	66.1	6
CBD-E-FPN [10]	ResNet101	54.2	77.0	71.5	87.1	44.6	75.4	63.5	76.2	65.3	79.3	79.5	47.5	59.3	69.1	69.7	64.3	84.5	59.4	44.7	83.1	67.8	-
CSFF [11]	ResNet101	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	81.6	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	68.0	68.0	10
One-stage methods																							
YOLOv3 [48]	DarkNet53	72.2	29.2	74.0	78.6	31.2	69.7	26.9	48.6	54.4	31.1	61.1	44.9	49.7	87.4	70.6	68.7	87.3	29.4	48.3	78.7	57.1	28
YOLOv4 [49]	Dark53	71.3	51.2	66.5	86.9	33.2	72.7	45.4	55.9	47.4	65.6	60.6	56.0	51.8	82.5	63.8	62.0	80.3	52.5	42.4	73.1	61.1	**43**
RetinaNet [27]	ResNet101	53.3	77.0	69.3	85.0	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5	66.1	12
FoveaBox [50]	ResNet50	66.9	79.6	**76.7**	87.6	42.7	79.8	60.6	81.1	66.4	74.5	80.3	50.0	57.6	73.1	71.5	60.0	86.9	54.5	42.7	**88.3**	69.0	-
FCOS [26]	ResNet50	61.1	82.6	76.6	87.6	42.8	**80.6**	64.1	79.1	67.2	**82.0**	79.6	46.4	57.8	72.1	64.8	63.4	85.2	62.8	43.8	87.5	69.4	-
SCRDet++ [13]	ResNet50	64.3	79.0	73.2	85.7	45.8	76.0	**68.4**	79.3	68.9	77.7	77.9	56.7	**62.2**	70.4	67.7	60.4	80.9	**63.7**	44.4	84.6	69.4	-
HawkNet [12]	ResNet50	65.7	**84.2**	76.1	87.4	45.3	79.0	64.5	**82.8**	72.4	50.2	**82.5**	**74.7**	59.6	**89.7**	66.0	**70.8**	87.2	61.4	**52.8**	88.2	**72.0**	10
Our methods																							
SSD*	VGG16	73.0	69.9	71.0	86.8	30.4	78.3	55.4	67.1	49.5	78.0	69.2	57.0	53.7	62.1	75.1	40.1	85.2	54.1	26.8	68.9	62.6	30
ASSD	VGG16	**85.6**	82.4	75.8	**89.5**	40.7	77.6	64.7	67.1	61.7	80.8	78.6	62.0	58.0	84.9	**76.7**	65.3	**87.9**	62.4	44.5	76.3	71.1	21
SSD-lite*	MobileV2	51.8	60.3	65.7	82.1	20.3	73.6	44.8	57.5	40.8	73.4	50.6	52.3	47.0	41.0	61.4	24.1	76.5	45.4	15.3	40.1	51.2	**43**
ASSD-lite	MobileV2	73.7	75.7	69.5	85.4	27.8	74.6	59.2	61.9	49.0	76.7	72.2	61.0	50.5	75.6	75.8	49.7	82.5	56.5	31.3	57.2	63.3	35

TABLE IV

ABLATION EXPERIMENTS ON THE DOTA DATASET

	SSD				ASSD
DA ?		√	√	√	√
SAA ?			√	√	√
PAPM ?				√	√
CFAM ?					√
mAP_{50} (%)	64.99	66.57	72.36	74.02	**76.01**

adapt to small regions and hence lacks enough information to utilize. While the FAM equipped with the CM can compensate for this weakness by the contextual information, thus the AP_N improvement increases as the size decrease like the aqua curve in Fig. 11(c).

e) Effect of SAA: The Scale SAA (SAA for short) is proposed to tackle the scale mismatch, and the overall performance is reported in Tables I and IV. Figs. 9 and 11 report the accuracy improvement as well. The 3.38% and 5.79% overall accuracy mAP_{50} gain shows the effectiveness of our method. More specifically, the orange curve in Fig. 11(c) illustrates that the impact of the SAA is even for the objects with different aspect ratios. This is because the scale arrangement of the predefined anchor is also updated in addition to the core anchor. These two types of anchors both respond to all objects with various aspect ratios.

We find that the scale aligned anchor mainly contributes to the small/extra-small and middle object shown as the orange curve in the left of Fig. 11(c). The performance comparison of small "Ship" and "Storagetank" in Fig. 11(a) and (b) also shows this trend. The reason is that we set the parameter m in 3 to 3, the scale arrangement of the first three layers will be rectified, and the scales of feature and anchor are matched better while the latter layers keep unchanged.

f) Ablation Experments on DOTA Dataset: The quantitative results of ablation experiments on the DOTA [28] dataset are reported in Table IV. Our reimplemented baseline model

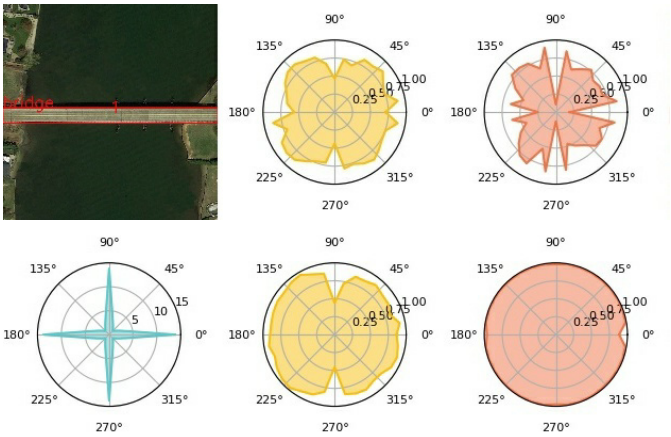


Fig. 12. Results of the robustness test. The first row shows the test image with a long-narrow "bridge," the location precision (yellow curve) and classification confidence (red curve) of the baseline model change as this object rotates from 0° to 360°. In the second row, the figure on the left plots the change of aspect ratio (blue curve) with the object rotation angle, and the second and last figure show our detector's results.

has an overall accuracy of 64.99%, which is relatively low compared to the state of the art (SOTA) results. To boost this performance, we add the data augmentation (DA for short) to it, in which the random flip and (90°, 180°, and 270°) random rotation are implemented. Besides, we also equip this model with our components to gain performance. Specifically, the SAA, PAPM, and CFAM bring 5.79%, 1.66%, and 1.99% overall mAP_{50} improvement, respectively. Finally, our method improves the baseline model by 9.44% mAP_{50} in total, which is also a proof of the effectiveness of our method.

2) Robustness Test: The proposed method aims to adaptively align anchor, object, and feature; hence, we can get a more robust feature extraction and target detection. To inspect our detector's robustness, we evaluate the ASSD on a long-narrow object with an extremely large aspect ratio, as shown in Fig. 12. We rotate the object "bridge" around

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TABLE V

ACCURACY (%) COMPARISON OF STATE-OF-THE-ART METHODS ON THE DOTA DATASET. "SSD*" MEANS OUR REIMPLEMENTATED BASELINE MODEL, "ASSD" IS OUR PROPOSED METHOD, "ASSD+" MEANS ASSD WITH MULTISCALE TESTING, AND "DARK53" MEANS CSPDARKNET-53

Method	Backbone	Pla	BD	Bri	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Har	SP	HC	mAP	FPS
Two-stage methods																		
ICN [3]	ResNet101	90.0	77.7	53.4	**73.3**	73.5	65.0	78.2	90.8	79.1	84.8	57.2	62.1	73.5	70.2	58.1	72.5	-
Yan [4]	ResNet101	88.6	80.2	53.2	67.0	76.3	72.6	84.1	90.7	81.0	76.2	57.1	66.7	**84.1**	66.4	56.9	72.7	1
SCRDet [5]	ResNet101	**90.2**	81.9	55.3	73.3	72.1	77.7	78.1	**90.9**	82.4	**86.4**	64.5	63.5	75.8	78.2	60.1	75.4	-
FADet [6]	ResNet101	90.2	78.6	51.9	75.2	73.6	71.3	81.4	90.9	83.9	84.8	58.9	65.7	76.9	79.4	68.2	75.4	-
SCRDet++ [13]	ResNet101	90.0	82.3	**61.9**	68.6	69.6	81.2	78.8	90.9	**86.3**	85.1	**65.1**	61.1	77.7	80.7	64.3	76.2	-
One-stage methods																		
YOLOv3 [48]	DarkNet53	79.0	77.1	33.9	68.1	52.8	52.2	49.8	89.9	74.8	59.2	55.5	49.0	61.5	55.9	41.7	60.0	13
SBL [1]	ResNet50	89.2	66.0	46.8	52.6	73.1	66.1	78.7	90.9	67.4	72.2	39.9	56.9	69.6	67.7	34.7	64.8	-
DSSD [51]	Hour104	91.1	71.8	54.6	66.4	79.0	77.2	87.5	87.6	52.1	69.7	38.0	**72.6**	75.4	59.4	28.9	67.4	9
YOLOv4 [49]	Dark53	88.8	78.1	48.3	56.0	70.7	77.4	77.8	90.6	76.8	81.1	41.4	64.9	76.2	75.1	56.3	70.7	**43**
FMSSD [7]	VGG16	89.1	81.5	48.2	67.9	69.2	73.6	76.9	90.7	82.7	73.3	52.7	67.5	72.4	80.6	60.2	72.4	16
EFR [8]	VGG16	88.4	83.9	45.8	67.2	76.8	77.2	85.4	90.8	85.6	75.8	54.6	60.8	71.4	77.9	60.9	73.5	-
SCRDet++ [13]	ResNet152	87.9	**84.6**	56.9	68.0	74.7	78.8	78.5	90.8	85.6	85.0	53.6	56.8	76.7	75.1	62.8	74.4	-
Our methods																		
SSD*	VGG16	87.1	75.4	30.0	57.7	66.2	70.9	76.8	90.8	75.4	55.9	48.2	50.2	62.5	69.2	58.6	65.0	30
ASSD	VGG16	89.3	82.0	52.1	62.6	**79.8**	81.8	88.2	90.9	82.0	82.9	59.8	65.3	77.1	**82.0**	64.5	76.0	21
ASSD+	VGG16	89.2	82.1	53.1	72.8	79.3	**82.7**	**88.3**	90.8	84.5	84.9	64.9	68.3	80.5	80.6	**64.6**	**77.8**	-



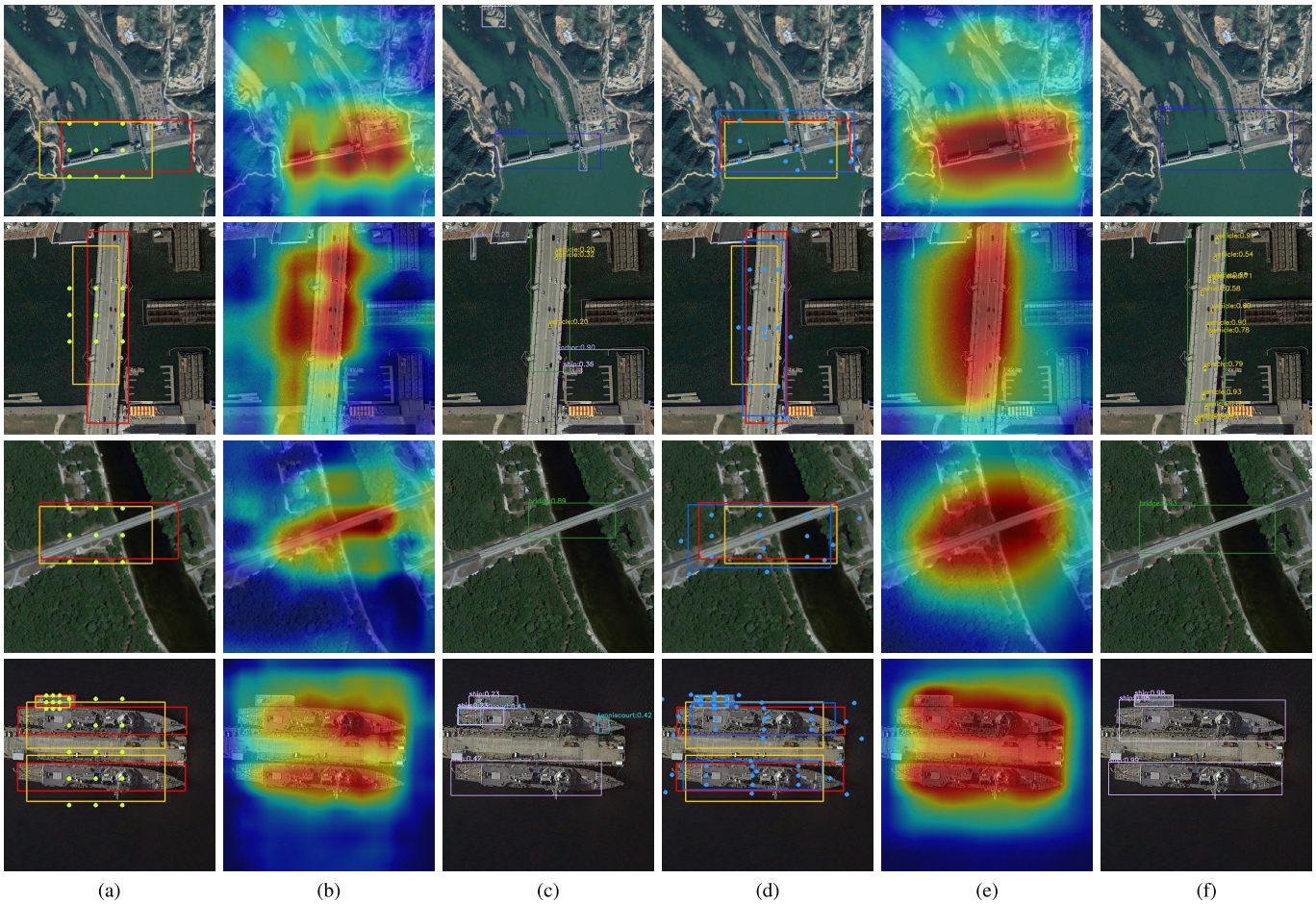
Fig. 13. Detection results of ASSD for the large-size remote sensing imagery. There are some long-narrow "harbor" targets and small "ship" targets in the DOTA dataset, which also shows the ability of ASSD to detect multiscale objects.

the image center from 0° to 360°. As the object rotates, the aspect ratio periodically changes from 15 to 1, as shown in Fig. 12. Meanwhile, the rotated images are fed into the baseline model and ASSD, and then, the location precision and classification confidence are plotted as the rotation angle increases. The results show that the baseline model outputs unstable classification confidence as the same object rotates, while the confidence of ASSD keeps high and stable. Besides, the proposed method improves the location precision by a large margin and maintains high robustness within a wide range of aspect ratios.

3) Comparison With the State of the Art: We compare the proposed ASSD with the state-of-the-art methods both on the DIOR [29] and DOTA [28] dataset. The results are detailed in Tables III and V, which shows that our detector ASSD achieves better balance between accuracy and speed than other models.

a) Results on DIOR: We compare ASSD with the state-of-the-art methods on the DIOR dataset in Table III, and the results of these involved methods are reported in [29] and related publications. The latest cross-scale feature fusion for object detection (CSFF) [11] and contextual bidirectional enhancement (CBD)-E-feature pyramid network (FPN) [10] achieves the top overall accuracy among the two-stage methods, but they are lower than our detector by 3.1% mAP and 3.3% mAP, respectively. The recent study shows that the one-stage detector can also acquire competitive performance compared to the two-stage methods. For example, the

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(a) (b) (c) (d) (e) (f)

Fig. 14. Comparison of the detection results on the DIOR dataset. (a) and (d) Ground truth (red rectangle), predefined anchors (yellow rectangle), aligned core anchors (blue rectangle), sampled points of standard convolution (yellow points), and CFAM (blue points). (b) and (c) CAM and detection results of the baseline model. (e) and (f) Corresponding results of ASSD. The CAM indicates which areas contribute to predicting a specific category. The hotter the region is, the more critical the contribution is.

small, cluttered and rotated objects detector (SCRDet)++ [13] gets 69.4% mAP and outperforms all the two-stage methods, while our ASSD is higher than it by 1.7% mAP. The accuracy of the latest single-stage HawkNet [12] is higher than ASSD by 0.9% mAP, while the speed is merely half of ours. The YOLOv4 [49] is almost twice as fast as ASSD, but the accuracy is lower by 10% mAP. Compared to our reproduced baseline SSD* [23], our method improves the overall accuracy by 8.5% mAP without bells and whistles. Compared with these advanced methods, our ASSD achieves the state-of-the-art performance of 71.1% mAP at a high speed of 21 FPS. The detection results are shown in Fig. 7. Besides, we replace the backbone network with the lightweight MobileNetV2 [52], and the performance gain shows that our method is robust to the backbone network. Finally, the "ASSD-lite" achieves a better tradeoff between accuracy and speed.

b) Results on DOTA: We compare the results of different methods on the DOTA dataset in Table V, and the results are reported on the official evaluation website [28] and the cited studies. On the whole, the two-stage methods achieve better overall accuracy, noting that the backbones of them are the heavy ResNet101 [53]/ResNeXt101 [54]. The one-stage detectors, such as the FMSSD [7] and enhanced feature representation (EFR) [8], have much lower accuracy than our ASSD with the same backbone VGG16. The latest SCRDet++ [13] achieves the best overall accuracy among the single-stage detectors, while our ASSD outperforms it by 1.6% mAP. YOLOv4 [49] is faster than ASSD, but the accuracy is relatively lower. Compared to our implementation of the baseline model SSD*, our method improves the overall accuracy by 11.0% mAP. Also, the final model with multi-scale augmentation achieves the state-of-the-art performance of 77.77% mAP, which outperforms all of these representative methods with heavy backbone. The detection results of each category are shown in Fig. 8, and the detection results on large-size remote sensing images and multiscale objects are shown in Fig. 13.

E. Qualitative Analysis

Fig. 10 compares the predefined anchors and the aligned core anchors. The results show that the aligned core anchors produced by the PAPM are aligned with objects better than the predefined anchors in spatial. The aligned core anchors increase the number of positive samples that can alleviate the neg/pos imbalance.

Fig. 9 plots the performance curves on the validation set in each ablation experiment. The results show that the PAPM could make the model training process more stable.

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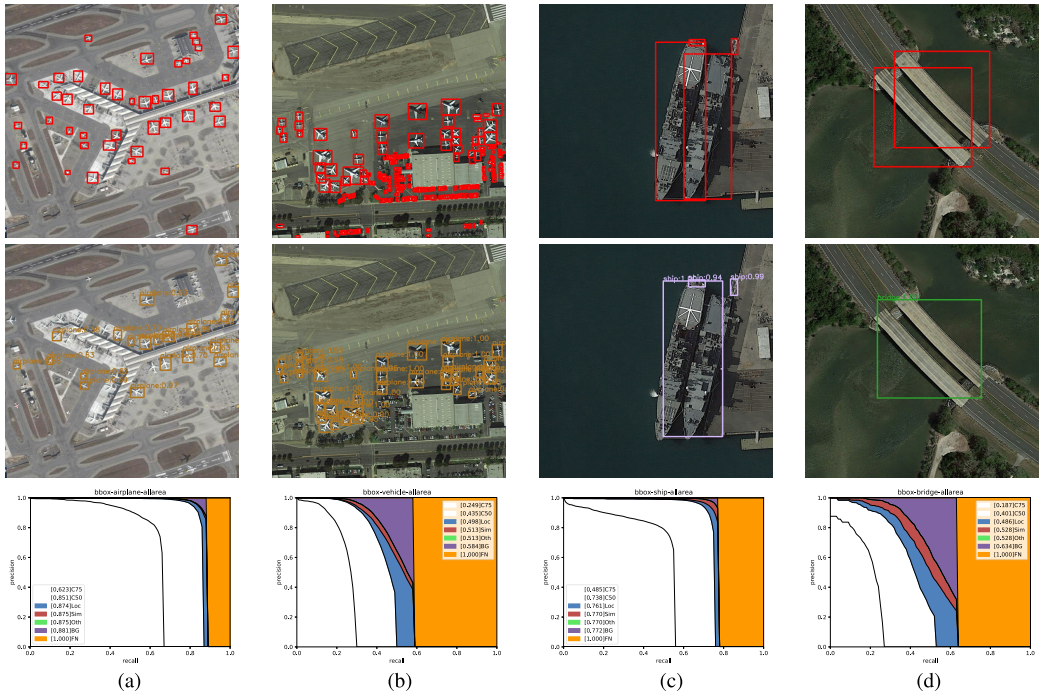


Fig. 15. False detection of ASSD on the DIOR dataset. The first row shows the ground-truth bounding boxes (red rectangle) of each category. The second row shows the false detection results. The last row shows the error analysis of the corresponding category. (a) and (b) FN detection of extremely small objects. (c) and (d) FN detection on the densely packed orientated objects with large overlaps. (a) Airplane. (b) Vehicle. (c) Ship. (d) Bridge.

The main reason is that the core anchors learn the target with top matching IoU within each anchor group, and the neg/pos imbalance is alleviated. Based on the aligned core anchors, the FAM and CFAM significantly improve the overall performance, as shown in this figure. The results also show the effectiveness of each component.

In Fig. 14, the detection results and the Class-Activation-Map (CAM) [55] of the baseline model and ASSD are compared. As shown in Fig. 14(a) and (d), the core anchors have more precise localization than the predefined anchors, and the sampled feature points in the convolution kernel are more flexible and more aligned to the corresponding objects. The CAM in Fig. 14(b) and (e) shows that the receptive fields of the baseline model are not aligned with objects. After equipped with the proposed alignment components, the detector can effectively address the feature misalignment problem, and the receptive fields adaptively focus on the entire region of each object. As a result, our detector has more precise localization and higher confidence than the baseline model. Our method significantly improves the detection of long-narrow objects, such as the bridge, ship, and airport. In a word, the misaligned feature causes misaligned detection and vice versa.

More results are shown in Fig. 13, we can find that the proposed ASSD can effectively detect long-narrow objects with extreme aspect ratio. Meanwhile, it can also detect multiscale geospatial objects, such as the extremely large "harbor" and densely packed small "ship."

F. Discussion

The typical false detection of ASSD is shown in Fig. 15. The results show that ASSD performs not very well on tiny objects and densely packed orientated objects.

According to the error analysis of the tiny "airplane" and "vehicle" objects, the main detection error comes from the false positive on the background (BG) and the false negative (FN). The main reason may be the absence of extremely tiny anchors and the large feature steps. Meanwhile, the CFAM samples feature points according to the corresponding object's shape; hence, the collected features lack adequate information to be correctly classified. Future work could enlarge the input image to ensure that small objects can be detected better.

Another typical false detection comes from the densely packed orientated objects. For the "ship" and "bridge" categories, the main errors are the "Loc" and "FN." We find that these objects are arranged so densely that only the detection with top confidence score has remained after the NMS operation. According to the study of [56]–[58], future work could detect with the orientated minimal bounding box instead of the present axis-aligned rectangle.

For future work, we will extend our alignment method to solve the misalignment problem in detecting orientated geospatial objects and speed up the proposed detector.

V. Conclusion

In this article, we explore the misalignment problem in one-stage detectors, which significantly limits the detection accuracy for geospatial objects of varied shapes and aspect ratios. This problem is summarized as the misaligned anchor and misaligned convolutional feature. We present a one-stage detection framework ASSD to address the misalignment problem, in which the PAPM is proposed to align anchor with objects in a novel way, and the CFAM is designed to adaptively align features with various objects in spatial so as to acquire robust and consistent features. As a result, the proposed method can rectify the misalignment of one-stage detectors.

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Extensive experiments demonstrate that ASSD achieves state-of-art performance on the DIOR and DOTA benchmarks.

ACKNOWLEDGMENT

The authors would like to thank all the reviewers for their valuable comments and feedback to improve the quality of the manuscript.

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